

Per-Round Parity Swap for Mirror RB with NewSampler

Overview

This document describes a targeted fix to `mirror_rb_experiment.py` that corrects a parity inconsistency in how entangling (two-qubit) gates are assigned across rounds when using the `NewSampler` algorithm with an odd number of circuit lengths.

File modified: `qiskit_device_benchmarking/bench_code/mrb/mirror_rb_experiment.py`

1. Problem Statement

`NewSampler` constructs Clifford layers with strict alternation: even-indexed layers contain only single-qubit (1q) gates, while odd-indexed layers contain the full two-qubit (2q) matching. When `full_sampling` is off, all circuit lengths within a sample are derived from the longest sequence by truncation. A subsequent reordering step (lines 381--392) reverses the Clifford layer order so that the deepest layer ends up at index 1 — the position used to determine entangling pairs and inject the entangling angle.

Whether this outermost layer is 1q or 2q depends on the circuit length:

Circuit length	Number of Clifford layers	Deepest layer index	Layer type
2	1	0	1q (even)
4	2	1	2q (odd)
10	5	4	1q (even)
20	10	9	2q (odd)
50	25	24	1q (even)

Within each sample of five lengths, the pattern alternates correctly (empty, full, empty, full, empty). However, with five lengths (an odd count), the pattern **inverts at every sample boundary**. For example, round 4 (length=50, empty) is immediately followed by round 5 (length=2, empty), producing two consecutive rounds without entangling pairs. This inconsistency means that downstream analysis observing "odd rounds should carry 2q pairs" cannot rely on that invariant across the full experiment.

2. Solution

After the existing Clifford reordering block, we insert a **per-circuit conditional swap** of the Clifford layers at indices 1 and 3 (together with their mirror counterparts) whenever the layer at index 1 does not match the desired round parity.

Logic

For each sequence indexed by `s`:

1. **Skip short circuits.** If the sequence has fewer than 9 elements (i.e., fewer than 2 Clifford layers), no swap is possible.
2. **Determine current state.** Check whether `sequence[1]` contains any 2q gates (`has_2q`).
3. **Determine desired state.** Odd rounds should carry 2q pairs: `want_2q = (s % 2 == 1)`.
4. **Swap if mismatched.** If `has_2q != want_2q`, swap the Clifford at index 1 with the one at index 3, and swap their mirror counterparts at indices `n-2` and `n-4` (where `n = len(sequence)`).

Code

The following block was added at line 394, immediately after the reordering step:

```
# Parity-aware swap: ensure outermost Clifford matches round parity
# (odd rounds -> 2q layer at index 1, even rounds -> 1q layer)
if not self.experiment_options.full_sampling and any(self._angles):
    for s, sequence in enumerate(sequences):
        if len(sequence) < 9: # need at least 2 Clifford layers
            continue
        has_2q = any(len(g.qargs) == 2 for g in sequence[1])
        want_2q = (s % 2 == 1)
        if has_2q != want_2q:
            n = len(sequence)
            # Swap outermost pair of Cliffords (indices 1<->3 and mirror
            # n-2<->n-4)
            sequence[1], sequence[3] = sequence[3], sequence[1]
            sequence[n - 2], sequence[n - 4] = sequence[n - 4], sequence[n
            - 2]
```

No changes were made to the `_pairs/_singles` extraction (lines 408--418) or the entangling angle injection (lines 461--471). Both continue referencing indices 1 and `n-2`, which now consistently hold the intended layer type after the swap.

3. Why the Mirror Structure Is Preserved

A mirror RB circuit of length `L` (with `k` Clifford layers) has the form:

$$P_0, C_{\{k-1\}}, P_1, C_{\{k-2\}}, \dots, C_0, P_k, C_0^{-1}, \dots, C_{\{k-2\}}^{-1}, P_{\{2k-1\}}, C_{\{k-1\}}^{-1}, P_{\{2k\}}$$

The forward half composes to $C_{\{k-1\}} * C_{\{k-2\}} * \dots * C_0$, and the inverse half undoes it as $C_0^{-1} * \dots * C_{\{k-2\}}^{-1} * C_{\{k-1\}}^{-1}$. The circuit implements the identity (up to Pauli corrections from the `P` layers).

The swap exchanges the Cliffords at indices 1 and 3 in the forward half, and their counterparts at indices `n-2` and `n-4` in the inverse half. Taking a length-10 circuit as an example (indices 0 through 20):

Before swap:

```
P0, C4, P1, C3, P2, C2, P3, C1, P4, C0, P5, C0-1, P6, C1-1, P7, C2-1,
P8, C3-1, P9, C4-1, P10
```

After swap (indices 1<->3, 19<->17):

```
P0, C3, P1, C4, P2, C2, P3, C1, P4, C0, P5, C0-1, P6, C1-1, P7, C2-1,
P8, C4-1, P9, C3-1, P10
```

The forward product becomes $C3 * C4 * C2 * C1 * C0$, and the inverse becomes $C0^{-1} * C1^{-1} * C2^{-1} * C4^{-1} * C3^{-1}$, which is still the exact inverse. The mirror identity is preserved.

This is consistent with the framework of Proctor et al. (Supplementary Note 5), where each layer in the circuit is independently sampled with its own random pairings. The success probability factorises per layer (Eq. 54), so reordering which independently sampled layer occupies a given position does not affect the protocol's validity.

4. Backward Compatibility

`edge_grab` and `matching` samplers

These samplers generate $2q$ gates in every Clifford layer. After reordering, index 1 always contains $2q$ gates regardless of circuit length. On odd rounds, `want_2q = True` and `has_2q = True`, so no swap occurs. On even rounds, `want_2q = False` and `has_2q = True`, so a swap is performed — but since both layers at indices 1 and 3 contain $2q$ gates (with different random pairings), the swap is statistically equivalent. No behavioural change.

Standard Mirror RB (non-QA)

The `_pairs` metadata is stored but is not consumed by `MirrorRBAnalysis`. The swap has no impact on standard Mirror RB workflows.

`full_sampling` mode

The swap block is guarded by `not self.experiment_options.full_sampling`, so it is never executed when full sampling is enabled.

5. Known Limitation: Length-2 Circuits

A length-2 circuit contains exactly one Clifford layer (layer 0), which is always 1q-only by `NewSampler`'s even/odd alternation rule. The resulting sequence has 5 elements:

```
P0, C0, P1, C0-1, P2
```

There is no second Clifford layer to swap with, so the guard `len(sequence) < 9` correctly skips these circuits. On odd rounds where `want_2q = True`, length-2 circuits will still report empty pairs.

Recommendation: When using `NewSampler` for Quantum Awesomeness experiments, either exclude length=2 from the length list, or use an even number of lengths (e.g., `[4, 10, 20, 50]`) so that odd rounds never coincide with length=2 at sample boundaries.

6. Verification

The fix was verified by running Cell 9 of `common.ipynb` with `lengths=[2, 4, 10, 20, 50]`, `num_samples=20`, `seed=123`, using the Aer stabilizer simulator with depolarizing noise (`p2 = 0.01`, `p1 = 0.001`).

NewSampler (exp03) — Parity Pattern

Round	Length	Round parity	Pairs	Status
0	2	even	empty	correct
1	4	odd	8 pairs	correct
2	10	even	empty	correct
3	20	odd	8 pairs	correct
4	50	even	empty	correct
5	2	odd	empty	correct (length-2 limitation)
6	4	even	empty	correct
7	10	odd	8 pairs	correct
8	20	even	empty	correct
9	50	odd	8 pairs	correct

This pattern holds consistently across all 100 rounds (20 samples). Before the fix, the pattern inverted at every sample boundary (e.g., rounds 4 and 5 were both empty, then round 6 was full).

Regression Check

- `edge_grab (exp01)`: All 100 rounds report non-empty pairs. No regression.
- `matching (exp02)`: All 100 rounds report 8 pairs each. No regression.

Circuit Validity

All experiments executed successfully on the Aer stabilizer backend, confirming that target bitstrings remain valid and the mirror identity is preserved after the swap.